Why genomic prediction of colistin resistance does not transfer between *Klebsiella pneumoniae* collections

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25 October 2026

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Code and data: https://github.com/omarfaruqe/klebsiella-amr

Abstract

Background. Genomic prediction of antimicrobial susceptibility is accurate for drugs whose resistance is conferred by acquired genes, but poor for colistin, where resistance in *K. pneumoniae* arises largely from loss of the chromosomal regulator *mgrB*. Why it is poor, and whether machine learning closes the gap, has not been examined at the level of individual mechanisms.

Methods. We assembled 6,652 public *K. pneumoniae* genomes from BV-BRC passing assembly quality control and carrying laboratory susceptibility results, and re-interpreted every minimum inhibitory concentration (MIC) under a single standard (EUCAST v16.1). Genotypes were called with AMRFinderPlus 4.2.7 and Kleborate 3.2.4. We added a targeted *mgrB* integrity check — tblastn of the reference MgrB protein against each assembly, classifying the locus as intact, missense, or disrupted — and benchmarked rule-based prediction against logistic regression and gradient boosting under random, study-aware (BioProject-grouped), lineage-aware (sequence-type-grouped) and temporal splits, with paired bootstrap comparisons. The rule was then applied, frozen, to an external cohort assembled from NCBI Pathogen Detection and de-overlapped against our own.

Results. Among 1,528 genomes with broth-microdilution colistin labels (367 resistant), *mgrB* was disrupted in 153 resistant versus 41 susceptible genomes. Adding mechanism-specified *mgrB* disruption raised colistin sensitivity from 28.3% (95% CI 24.0–33.2) to 56.7% (51.6–61.7) for AMRFinderPlus and from 39.5% (34.6–44.6) to 45.5% (40.5–50.6) for Kleborate, which already calls MgrB truncations. The association was not an artefact of assembly fragmentation: within the quartile of least fragmented assemblies, 81% of *mgrB*-disrupted genomes were resistant against 4% of intact ones, and the gap persisted across all quartiles. The breaks carried insertion-sequence signatures: they lay a median of 79 bp from *mgrB* against 80,470 bp in genomes with intact *mgrB*, the adjacent sequence was multi-copy within its own assembly more often than length-matched control contig ends (72% versus 35% with three or more copies), and 98% of these sequences recurred in unrelated isolates. Crucially, leave-one-study-out analysis showed sensitivity of this unfitted rule ranging from 39.5% to 70.6% between collections; in the largest single study *mgrB* disruption

accounted for under 40% of colistin resistance. A second tier based on *mgrB* missense variants appeared to add 19.1 points of sensitivity in-sample but 0.9 points when variants were selected out-of-sample, and was discarded. Against machine learning, the ordering inverted with validation stringency: models exceeded the rule under random splitting (balanced accuracy 0.844 and 0.842 versus 0.759; paired differences −0.085 and −0.083, both excluding zero) and under study-aware splitting, but fell below it under lineage-aware splitting (0.731 and 0.709 versus 0.759; +0.028, 95% CI +0.002 to +0.052, and +0.050, +0.026 to +0.075). Correcting across the family of 54 comparisons, the reversal against gradient boosting survives both Benjamini–Hochberg and Bonferroni (q = 0.001) while that against logistic regression survives only the former (q = 0.045), so the latter is better read as parity. A predictor using sequence type alone reached 0.737 for meropenem but only 0.554 for colistin, and study-aware and lineage-aware splits each left the other dimension 74–95% overlapping. On 263 external isolates not in our cohort, the frozen rule reached 77.8% sensitivity (67.6–85.5) at 99.5% specificity; the *mgrB* component performed as it did internally (45.7% versus 41.7%) while AMRFinderPlus alone rose from 28.3% to 50.6%, because external resistance was weighted toward chromosomal point mutations in its catalogue rather than the insertion-driven disruption it misses. Six of 270 NCBI-labelled *K. pneumoniae* were *K. aerogenes*.

Conclusions. The apparent superiority of machine learning for colistin reverses when lineages are held out. It is not explained by lineage labels themselves: a sequence-type-only predictor reached just 0.554, so what the models lose is a finer-grained resemblance between related isolates. More fundamentally, the mechanistic composition of colistin resistance differs substantially between public genome collections — a limit that random and even study-aware validation conceals. External validation bears this out in both directions: the targeted *mgrB* check replicated, while the same rule's overall sensitivity differed by 21 points between collections for reasons of composition alone. We recommend that genomic AMR evaluations report a mechanism-specified rule baseline, harmonise breakpoints explicitly, and use lineage-aware splits.

Keywords: antimicrobial resistance, *Klebsiella pneumoniae*, colistin, *mgrB*, genomic prediction, external validity, data leakage

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1. Introduction

For most drug–organism combinations in *Klebsiella pneumoniae*, resistance is conferred by acquired genes or well-characterised target mutations, and tools such as AMRFinderPlus and Kleborate detect them reliably. Colistin is the conspicuous exception, and the exception matters: colistin is a last-line agent against carbapenem-resistant *K. pneumoniae*, and resistance arises predominantly not from an acquired gene but from loss of function of *mgrB*, a 47-codon chromosomal gene encoding a negative regulator of the PhoPQ two-component system. Loss of MgrB de-represses PhoPQ, modifying lipid A and reducing colistin binding.

The commonest route to *mgrB* loss is insertion-sequence (IS) transposition into the gene, which creates a specific detection problem. An IS element inserted into a 144-base-pair coding sequence is typically longer than the repeat a short-read assembler can resolve, so the assembly breaks inside *mgrB*. A gene-presence screen finds no acquired colistin determinant; a point-mutation screen finds no codon change, because the alignment simply ends. The genome appears wild type at the one locus that matters.

Two further issues complicate measuring how well genotype predicts phenotype on public data. Deposited labels mix interpretive standards, so identical MICs can be reported susceptible under one CLSI revision and resistant under EUCAST, introducing systematic noise easily mistaken for genotype error. And public collections are dominated by outbreak investigations and surveys of high-risk clones, so closely related isolates are abundant and

random train/test splitting places near-identical genomes on both sides. Yu and colleagues recently demonstrated across more than 24,000 genomes from five species that models trained this way substantially conflate lineage markers with resistance determinants, and recommended comparison against rule-based baselines as future work [1].

We take that recommendation as the design principle and ask a mechanism-level question: when a resistance determinant is detectable but routinely missed, does supplying it to a rule outperform letting a model find it, and does either transfer to new collections? We assemble a 6,652-genome cohort with harmonised labels, add a targeted *mgrB* check designed around the assembly-break failure mode, and evaluate rules and models under progressively stricter validation.

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2. Methods

2.1 Cohort assembly and provenance

All *K. pneumoniae* (taxon 573) AMR phenotype records were downloaded from the BV-BRC `genome_amr` endpoint (snapshot 2026-10-22; 1,840,793 records, 39,598 genomes). Retrieval re-run the following day reproduced byte-identical output (SHA-256 d0e561a6…1505126). Every data pull writes a manifest recording query, timestamp, record count and checksum.

Only records with `evidence = "Laboratory Method"` were used as labels (85,291 records, 7,276 genomes); computational predictions deposited in the same database were excluded as circular.

2.2 Label harmonisation

Labels follow EUCAST Clinical Breakpoint Tables v16.1 [2]. Each record carrying an MIC was re-interpreted (S if MIC ≤ s_le, R if MIC > r_gt, else I) rather than accepting the deposited call. Censored measurements that cannot be resolved against the breakpoint yield no label.

This recovered substantial data: records deposited without an S/I/R call were uninterpreted measurements rather than missing data, and re-interpretation added 1,921 labelled genomes for meropenem, 959 for ciprofloxacin, 598 for colistin and 560 for gentamicin. It also resolved most apparent contradictions — meropenem label conflicts fell from 60 genomes to 8 — because the majority were the same genome interpreted under different standards by different studies. Where both a deposited and a re-interpreted call existed they agreed in 92.9–97.2% of records.

Three policies were fixed in advance and recorded as decision records in the repository: intermediate and conflicting labels are excluded; colistin labels are restricted to broth microdilution, the only method EUCAST accepts for colistin MICs; and study-level label sets judged implausible on review are flagged and excluded with evidence recorded (one flag applied, §3.9). Tigecycline has no EUCAST breakpoint for *K. pneumoniae*, so deposited calls were used.

2.3 Assembly quality control and genotyping

Assemblies were required to be *K. pneumoniae* by both species field and genome name, 5.0–6.5 Mb, ≤500 contigs, N50 ≥20 kb, CheckM completeness ≥95% and contamination ≤5%. Of 6,817 labelled genomes 6,652 passed. Requiring agreement between species field and genome name removed four genomes with internally inconsistent BV-BRC records, one independently confirmed as *Citrobacter freundii*.

Assemblies were downloaded from BV-BRC by genome identifier — 63% of QC-passing genomes (4,190/6,652) have no NCBI assembly accession — and validated for format and length. AMRFinderPlus 4.2.7 (database 2026-08-07.1) was run in nucleotide mode with `--organism Klebsiella_pneumoniae --plus`, and Kleborate 3.2.4 with Kaptive 3.2.2 using the `kpsc` preset. All 6,652 genomes completed both tools (8.8 h wall time, ~140 CPU-hours, zero failures). A cross-machine verification script reproduces identical calls on macOS arm64 and Ubuntu x86_64.

2.4 Rule-based prediction

Per-drug rules were specified from tool output before phenotype comparison: acquired carbapenemase for meropenem; QRDR mutation or acquired quinolone gene for ciprofloxacin; aminoglycoside-modifying enzyme or 16S rRNA methyltransferase for gentamicin; *mcr* or *mgrB*/*pmrAB*/*phoPQ* determinants for colistin; acquired *tet(X)* or *tmexCD-toprJ* for tigecycline.

Two corrections were applied after a 524-genome pilot, both tool-annotation artefacts rather than biology. AMRFinderPlus assigns the intrinsic chromosomal *oqxAB* pump to tigecycline as well as quinolones; since essentially all *K. pneumoniae* carry *oqxAB* this produced 88 of 91 pilot tigecycline false positives, and *oqxAB* is now excluded for all drugs. Efflux-regulator variants (*oqxR*, *ramR*, *acrR*) produced 21 of 26 pilot ciprofloxacin false positives and are reported as a separate tier rather than folded into the core rule.

2.5 Targeted *mgrB* integrity check

The MgrB reference protein (WP_002911375.1, 47 aa) was searched against each assembly with tblastn (e-value ≤1e-5, hits ≥10 aa) and the best hit classified as **intact** (full-length, no residue differences); **missense** (full-length with substitutions); **premature_stop**; **frameshift_or_indel** (two fragments on one contig <300 bp apart); **insertion_assembled** (two fragments >300 bp apart); **split_contigs**; **contig_break** (alignment terminating at a contig edge); **partial**; or **absent**. All categories except intact and missense count as disrupted. The `split_contigs`, `contig_break` and `partial` categories are the assembly signature described in §1. The search takes 2 minutes for 6,652 assemblies on 16 threads.

Residue differences are enumerated by direct comparison of aligned sequences, ignoring query position 1, because tblastn translates alternative start codons (GTG, TTG) as V or L rather than M. An initial implementation using percentage identity classified 6,048 genomes as missense at exactly 97.9% identity — a single artefactual mismatch at the start codon.

2.6 Testing the insertion-sequence interpretation

The `split_contigs`, `contig_break` and `partial` categories are inferences from assembly structure, so we tested three predictions that follow from IS insertion but not from incidental fragmentation. First, position: the break should coincide with *mgrB*, compared against genomes with intact *mgrB*. Second, repetitiveness: sequence at the *mgrB*-adjacent contig end should have copies elsewhere in the same assembly. Because contig ends are repeat-enriched in general, this was compared against randomly chosen other contig ends in the same assemblies, truncated to the same length as each genome's *mgrB* flank. Third, recurrence: the same sequences should appear at *mgrB* in unrelated isolates. Flanks were taken as up to 600 bp of contig end adjacent to *mgrB* (minimum 50 bp) and compared by blastn at ≥95% identity (within genome) or ≥90% (between genomes) over ≥80% of the flank.

2.7 Validating the *mgrB* rule

Because three disruption categories are inferred from assembly structure, we tested whether the association with phenotype could reflect assembly fragmentation by compar-

ing contig counts and N50 between calls, and by stratifying the colistin association on assembly-quality quartile.

Because the decision to treat missense variants as resistance-predicting was made after observing their enrichment in this cohort, we re-estimated that tier by leave-one-study-out: variants were selected on training studies only (≥3 genomes, >50% resistant) and applied to the held-out study. The disruption tier, being mechanism-specified and fitted to nothing, was evaluated in the same framework for comparison.

2.8 Attributing performance to lineage

To test directly, rather than infer, how much performance is obtainable from lineage, we built a predictor using sequence type alone: for each test genome it returns the majority label of that genome's sequence type among training genomes, falling back to the overall training majority when the type is unseen. It uses no resistance determinant. We also measured, for each split scheme, the fraction of test genomes sharing a sequence type or a BioProject with any training genome.

2.9 Benchmark

Features were 226 binary indicators per genome: AMRFinderPlus gene families (allele suffixes collapsed, OXA-48-like kept distinct from other OXA enzymes), 100 point-mutation calls, 33 AMRFinderPlus functional subclasses, and two *mgrB* features. Features present in fewer than five genomes were dropped.

Three predictors were compared: the per-drug rule (for colistin including mechanism-specified *mgrB* disruption, but not the missense tier, for the reason given in §3.4); L2 logistic regression (C = 1, class-balanced); and gradient boosting (HistGradientBoostingClassifier, 200 iterations, 15 leaves, class-balanced). Hyperparameters were fixed rather than tuned, since the object of study is validation design rather than peak performance.

Four split schemes were used: random 5-fold; study-aware 5-fold grouped by BioProject; lineage-aware 5-fold grouped by sequence type; and temporal. Genomes with missing group metadata were assigned singleton groups so absent metadata can never merge unrelated isolates. Metrics are computed on pooled held-out predictions with 95% Wilson intervals, reported both on all test labels and with borderline-MIC labels removed. A split is scored only if its test set contains at least 10 resistant and 10 susceptible genomes. Differences between predictors were assessed by paired bootstrap over held-out genomes (2,000 resamples), resampling once per iteration and recomputing both predictors on the same resampled set.

Comparing every drug, split scheme and predictor pair yields a family of 54 tests, not one, so some intervals will exclude zero by chance. Each comparison therefore also carries a two-sided bootstrap p-value, floored at 1/2,000 since no finer resolution exists, and a Benjamini–Hochberg q-value computed across the whole family. Benjamini–Hochberg is our primary correction: the comparisons share predictors, genomes and folds and are therefore positively dependent, and controlling the false discovery rate is the appropriate goal for a benchmark mapping where differences exist rather than testing a single hypothesis. We also report which comparisons survive the far stricter Bonferroni threshold ($p < 0.00093$), and we treat only those as established beyond doubt.

2.10 External validation

Leave-one-study-out within BV-BRC holds out a study but not an aggregator: every such study was curated the same way and was available to us when the rule was specified. To test the rule on genomes whose phenotypes we had no access to, we assembled a cohort from NCBI Pathogen Detection, joining submitted antimicrobial susceptibility results (`pdbrowser.ast`) to isolate records (`pdbrowser.isolates`) for the assembly accession, since

neither table carries both the MIC and the accession. All *K. pneumoniae* colistin results with an assembly accession were retrieved (n = 355).

Because BV-BRC ingests from NCBI, overlap was removed before anything else: isolates matching our cohort by BioSample accession, or by assembly accession ignoring the version suffix, were excluded. Labels were re-derived from the submitted MIC under the same pinned EUCAST table used internally, never taken from the submitted phenotype; isolates without a usable MIC were retained as a separate stratum carrying the submitter's own interpretation. Assemblies were downloaded from NCBI Datasets and verified against each archive's md5. AMRFinderPlus and Kleborate were run through the identical pipeline code as the internal cohort, and the rules were imported unchanged from the analysis module. Nothing was refitted, reselected or thresholded for this cohort, and the evaluation exposes no parameter that could be.

2.11 Reproducibility

All analyses are deterministic for a fixed seed (20261024). Code, pinned environments, manifests, decision records, reports and figures are at https://github.com/omarfaruqe/klebsiella-amr.

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3. Results

3.1 Cohort

The cohort comprised 6,652 QC-passing genomes with harmonised labels: meropenem 5,541 (1,854 R / 3,687 S), ciprofloxacin 4,640 (3,441 / 1,199), gentamicin 4,760 (2,113 / 2,647), colistin 1,528 broth-microdilution (367 / 1,161), tigecycline 948 (261 / 687). It is structurally non-random, as expected: two BioProjects contribute 44% of genomes, the ten commonest sequence types account for 52%, and collection year is missing for 34% (2,266/6,652).

3.2 Known mechanisms explain four drugs well and colistin poorly

Drug	AMRFinderPlus sensitivity	specificity	Kleborate sensitivity	specificity
Ciprofloxacin	98.8% (98.4–99.1)	88.2%	98.7%	88.4%
Meropenem	96.7% (95.7–97.4)	91.2%	96.5%	91.3%
Gentamicin	85.1% (83.6–86.6)	94.9%	81.4%	95.8%
Colistin	**28.3% (24.0–33.2)**	97.8%	**39.5% (34.6–44.6)**	96.3%
Tigecycline	**2.7% (1.3–5.4)**	100.0%	**0.0%**	100.0%

The tools agree on 97.8–99.9% of genome–drug pairs for every drug except colistin (87.8%), the one drug where they differ in what they look for: Kleborate's higher sensitivity derives from MgrB truncation calls that AMRFinderPlus's point-mutation screen does not make. For colistin, 263 of 367 resistant genomes carried no AMRFinderPlus determinant at all.

3.3 A targeted *mgrB* check recovers part of the gap

Across all 6,652 genomes *mgrB* was intact in 6,056, missense in 129, and disrupted in 467, the last dominated by the assembly signatures predicted in §1: split across contigs (162), partial (153), premature stop (73), absent (34), frameshift or indel (27), contig break (13) and assembled internal insertion (5).

Among the 1,528 genomes with broth-microdilution labels (24% resistant overall):

mgrB status	Resistant	Susceptible
Disrupted	153	41
Missense	70	13
Intact	144	1,107

Adding mechanism-specified disruption to the rules (Figure 1):

Rule	Sensitivity (95% CI)	Specificity (95% CI)
AMRFinderPlus	28.3% (24.0–33.2)	97.8% (96.7–98.5)
AMRFinderPlus + *mgrB* disrupted	**56.7% (51.6–61.7)**	**95.2% (93.8–96.3)**
Kleborate	39.5% (34.6–44.6)	96.3% (95.0–97.2)
Kleborate + *mgrB* disrupted	**45.5% (40.5–50.6)**	**95.3% (94.0–96.4)**
mgrB disrupted alone	41.7% (36.8–46.8)	96.5% (95.2–97.4)

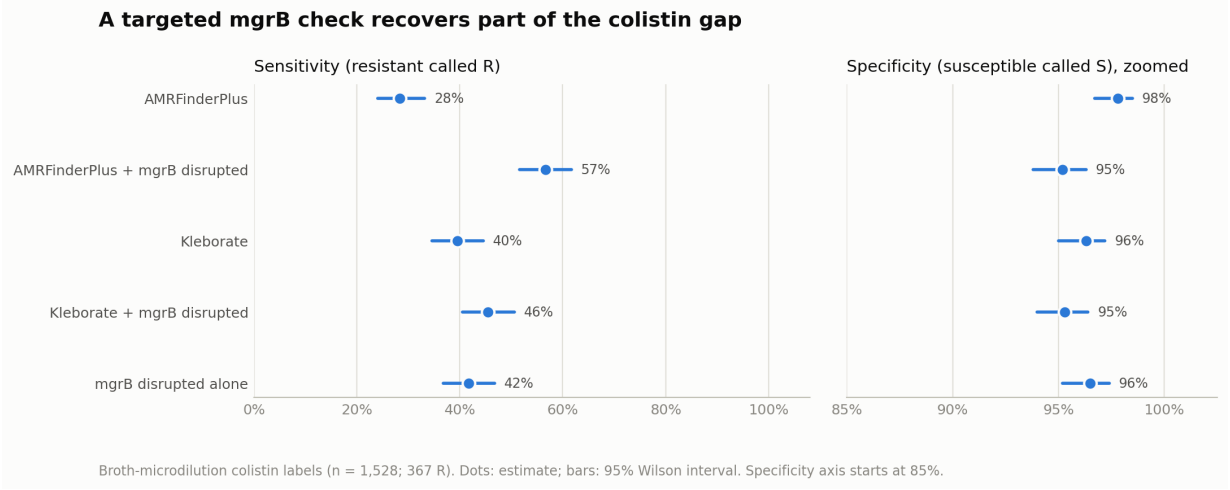


Figure 1. Colistin sensitivity and specificity for each rule, with and without the targeted *mgrB* disruption check. Dots are point estimates, bars 95% Wilson intervals; broth-microdilution labels (n = 1,528; 367 resistant). The specificity axis begins at 85%.

The increment is large against AMRFinderPlus (+28.4 points) and modest against Kleborate (+6.0 points), which already detects MgrB truncation by sequence coverage. Kleborate is the appropriate comparator for the state of the art, and the honest gain is therefore six points. Notably, the *mgrB* check used alone, with no gene screen, slightly exceeds Kleborate alone.

The association is not an assembly artefact. Median contig counts were similar between calls (disrupted 116, intact 104, missense 108), and stratifying on assembly quality left the association intact:

Contig quartile	*mgrB* intact, % resistant	*mgrB* disrupted, % resistant
Q1 (fewest contigs)	4% (n=337)	81% (n=32)
Q2	8% (n=315)	69% (n=48)
Q3	17% (n=292)	83% (n=65)
Q4 (most contigs)	18% (n=307)	82% (n=49)

The gap is widest, not narrowest, among the least fragmented assemblies. Disruption calls inferred from assembly structure behaved similarly to sequence-level calls (76% versus 86% resistant, against 12% for intact *mgrB*), indicating that the assembly-break categories carry genuine signal.

3.4 The contig breaks carry insertion-sequence signatures

Among the 328 genomes whose *mgrB* disruption was inferred from assembly structure, the break coincided with the gene: median distance from the *mgrB* fragment to the contig end was 79 bp, with 63% within 100 bp and 87% within 1 kb. In 328 matched genomes with intact *mgrB*, the gene lay a median of 80,470 bp from any contig end, and 0% were within 100 bp. Whatever terminates these assemblies is located at *mgrB* specifically.

The adjacent sequence behaved like a mobile element. Of 255 genomes with at least 50 bp of flank, 82% had a high-identity copy elsewhere in their own assembly and 72% had three or more (median 5), against 66% and 35% (median 2) for length-matched random contig ends from the same assemblies. And 249 of 255 flanks (98%) matched a flank from a different genome, which idiosyncratic assembly artefacts would not.

Together these support the interpretation that the assembly-inferred categories represent IS insertions in *mgrB*. They do not identify which element is involved, nor confirm insertion in any individual genome; both require an IS reference database or long-read assemblies of these isolates.

3.5 A plausible missense tier does not survive out-of-sample selection

Missense variants were enriched among resistant isolates — M27K (25 R / 3 S), T21N (9 / 0), N42Y (7 / 0), K43I (7 / 0), C28W (4 / 1), C39G (3 / 0) — and C28 and C39 are the two cysteines of this small lipoprotein, so a structural rationale was readily available. Counting missense as resistance-predicting raised in-sample sensitivity of the *mgrB* check from 41.7% to 60.8%, an apparent gain of 19.1 points.

Under leave-one-study-out selection the tier added **0.9 points** (47.0% to 47.9%). The apparent benefit was almost entirely an artefact of selecting variants on the same data used to evaluate them. We therefore exclude the missense tier from all reported rules and from the benchmark.

We report this negative result because the failure mode is instructive: the variants were individually plausible, the structural rationale was real, and the in-sample improvement was large and would have survived cross-validation that did not group by study.

3.6 Mechanism composition differs between collections

The disruption tier is fitted to nothing, so its leave-one-study-out performance measures transfer rather than overfitting. It varied substantially:

Held-out study	Resistant	Sensitivity, disruption only
PRJEB10018	162	39.5%
PRJNA504600	27	48.1%
PRJEB22890	63	50.8%
PRJNA288601	20	55.0%
PRJNA292904	7	57.1%
PRJNA307517	19	63.2%
PRJEB10561	17	70.6%
Pooled	**315**	**47.0% (41.5–52.5)**

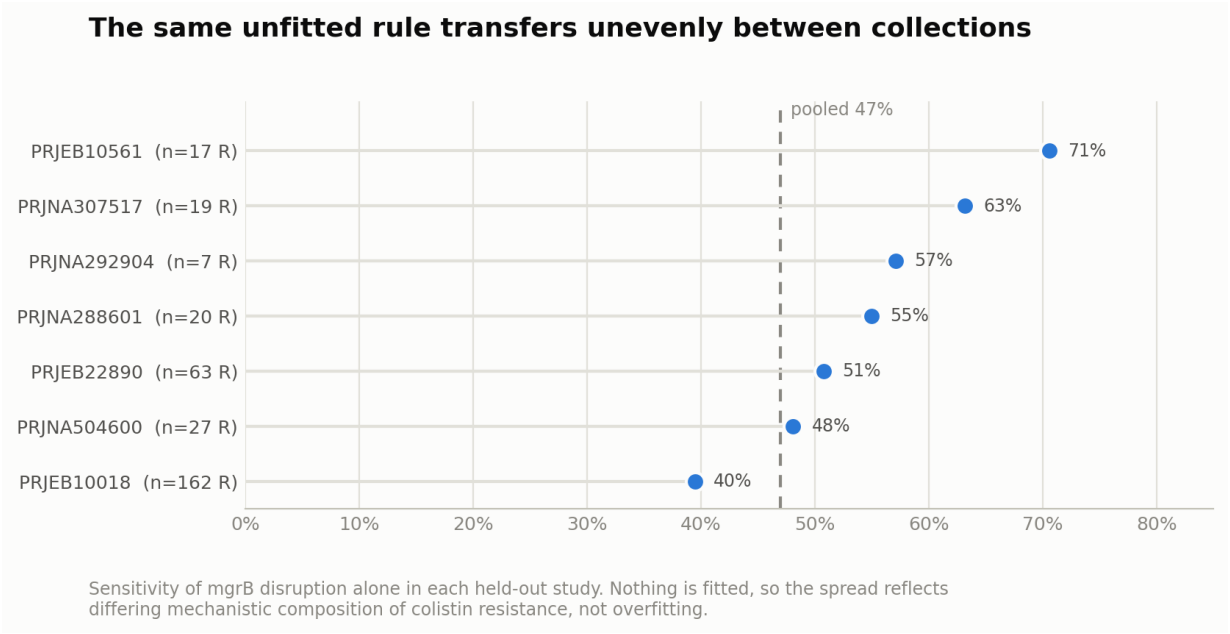


Figure 3. Sensitivity of the *mgrB* disruption rule in each held-out study. The rule is mechanism-specified and fitted to nothing, so the spread reflects differing mechanistic composition between collections rather than overfitting.

In the largest collection, *mgrB* disruption accounts for under 40% of colistin resistance; in another it accounts for over 70%. Since nothing is fitted, this is not an overfitting artefact but a statement about the isolates: the mechanistic composition of colistin resistance genuinely differs between collections, presumably reflecting differences in clonal background, geography and selection history. Any fixed rule, and any model trained on one collection, inherits this limit.

3.7 The ordering of rules and models inverts with validation stringency

Split	Rule + *mgrB*	Logistic regression	Gradient boosting
Random	0.759	**0.844**	**0.842**
Study-aware	0.759	**0.822**	**0.810**
Lineage-aware	**0.759**	0.731	0.709

(balanced accuracy on held-out genomes; Figure 2)

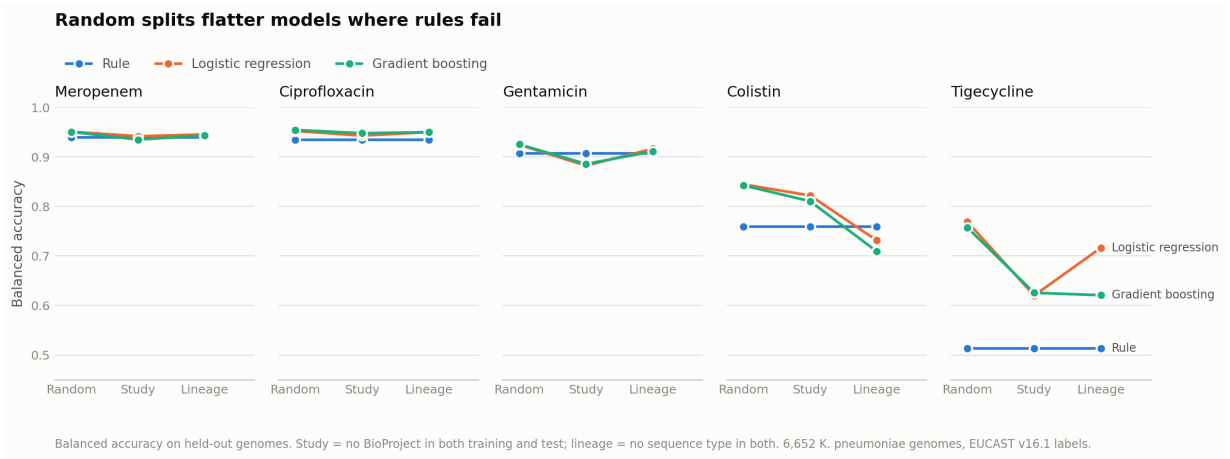


Figure 2. Balanced accuracy of the per-drug rule, logistic regression and gradient boosting under random, study-aware and lineage-aware splits. Rules are deterministic and therefore flat across splits by construction; the comparison of interest is whether model lines converge on, or fall below, the rule line as grouping tightens.

Paired bootstrap differences (rule minus model, 95% interval) confirm the reversal: under random splitting the models are ahead by 0.085 (−0.113 to −0.059) and 0.083 (−0.111 to −0.054); under study-aware splitting by 0.063 (−0.088 to −0.037) and 0.051 (−0.078 to −0.026); under lineage-aware splitting the rule is ahead by 0.028 (+0.002 to +0.052) and 0.050 (+0.026 to +0.075). All six intervals exclude zero, but they are drawn from a family of 54 comparisons and do not survive correction equally. The rule's margin over gradient boosting under lineage-aware splitting holds at $p < 0.0005$ (Benjamini–Hochberg $q = 0.001$) and survives Bonferroni; its margin over logistic regression does not ($p = 0.026$, $q = 0.045$, Bonferroni threshold 0.00093). The reversal against gradient boosting is therefore established, while the reversal against logistic regression should be read as comparable performance rather than superiority. Across the family, 33 of 54 intervals excluded zero uncorrected, 31 survived Benjamini–Hochberg and 24 survived Bonferroni; the two that fell away under Benjamini–Hochberg were small ciprofloxacin study-aware differences (−0.008 and −0.005) on which no claim here rests.

The models had access to the same *mgrB* features as the rule, and lost ground only when sequence types were held out. The margin by which the rule leads under lineage-aware splitting is small, and the practical claim is not that rules are superior but that the models' apparent superiority is an artefact of validation design.

Across drugs, random-split optimism — balanced accuracy under random splitting minus the lower of the two grouped splits — was 0.007–0.042 for meropenem, ciprofloxacin and gentamicin, but 0.113–0.133 for colistin and 0.136–0.149 for tigecycline. For tigecycline, models exceeded the rule under every split including lineage-aware (differences −0.203 and −0.107), though at 0.62–0.72 all predictors remain weak. Temporal validation was interpretable only for meropenem, ciprofloxacin and gentamicin; colistin (34 test genomes, one resistant) and tigecycline (28, none) were not evaluable.

3.8 The leaked signal is finer-grained than sequence type

A predictor using sequence type alone, with no resistance determinant of any kind, reached 0.737 balanced accuracy for meropenem, 0.728 for ciprofloxacin and 0.713 for gentamicin under random splitting, confirming that lineage carries substantial predictive signal in this cohort. For those three drugs the models nonetheless reached ~0.95 and did not degrade when sequence types were held out, so their performance rests on resistance determinants rather than lineage.

Colistin behaved differently, and not as we had expected. Sequence type alone reached only 0.554 under random splitting, despite 91% of test genomes sharing a type with training data. Colistin resistance is therefore poorly predicted by sequence type: both phenotypes occur within the same types. Yet the models still fell from 0.844 to 0.731 when types were held out. The signal they lose is thus not sequence-type membership as such, but something finer-grained that co-varies with it — plausibly accessory-genome similarity among closely related isolates within a type. We report this because it corrects a natural but incorrect reading of the split results: the models were not simply memorising lineage labels.

The two grouping schemes also proved not to be substitutes. Under study-aware splitting, 74–88% of test genomes still shared a sequence type with training data; under lineage-aware splitting, 87–95% still shared a BioProject. Each scheme leaves the other dimension almost entirely unconstrained, so grouping on both would be stricter than either alone. Temporal splitting, often presented as the realistic option, constrained neither: 73–78% of test genomes shared a sequence type and 76–82% a BioProject with training data.

3.9 Much apparent genotype error is label artefact

Meropenem. Of 2,118 carbapenemase carriers, 326 tested susceptible, strongly enzyme-dependent: 52.9% of OXA-48-like carriers (182/344) and 75.0% of VIM carriers (33/44) versus 5.9% of KPC (75/1,278) and 10.0% of NDM (28/280). Of the 240 with a recorded MIC, 191 (80%) lay at 1–2 mg/L and 224 (93%) exceeded the EUCAST carbapenemase screening cut-off of 0.125 mg/L. The genotype call is correct in most cases; the clinical S category does not reflect carbapenemase carriage for these families. Of 62 resistant non-carriers, 51 carried an ESBL or AmpC and 29 of those also had an OmpK35/36 defect — but 624 susceptible non-carriers had the same combination, making it a risk factor rather than a predictor.

Gentamicin. Of 299 unexplained resistant isolates with an MIC, 213 (71%) were at exactly 4 mg/L, one doubling dilution above the EUCAST breakpoint, whereas isolates with a detected mechanism were mostly ≥8 mg/L. Among CLSI-tested records in this borderline group, 160 of 162 had been reported susceptible by the originating laboratory. Excluding borderline labels raises gentamicin rule balanced accuracy from 0.906 to 0.959, above every model under every split.

A flagged study. Of 135 gentamicin false positives, 65 carried *rmtB*; 63 came from a single BioProject at MIC 1 mg/L, implausible for a 16S rRNA methyltransferase. That study's gentamicin labels are flagged and excluded, with evidence recorded.

3.10 The rule transfers to an external collection, which is composed differently

Of 355 NCBI colistin results, 84 (23.7%) were already in our cohort and were excluded, leaving 271 held out. Kleborate's species check rejected six that NCBI's taxgroup labels *K. pneumoniae*: all six are *Klebsiella aerogenes*, four of them complete single-contig genomes, so this is misassignment rather than poor assembly. AMRFinderPlus processed all six without complaint. A cohort built from NCBI taxgroups without an independent species check therefore inherits a 2.2% contamination rate from a source that does not flag it.

Of the 270 held-out isolates with a usable label, six were excluded as *K. aerogenes* and one intermediate isolate cannot inform sensitivity, leaving 263 (81 resistant, 182 susceptible). The frozen rule reached 77.8% sensitivity (95% CI 67.6–85.5) at 99.5% specificity, against 56.7% (51.6–61.7) internally; restricting to the 251 isolates with MIC-derived labels gave 78.3% (67.2–86.4). The rule therefore transfers, and the external estimate is higher than the internal one with non-overlapping intervals.

The reason is composition, not rule quality. The *mgrB* component performed almost identically in the two collections (45.7% versus 41.7% alone), while AMRFinderPlus alone rose from 28.3% to 50.6%. Among the 81 resistant isolates, 26 were explained by an AMRFinderPlus determinant alone, 22 by *mgrB* disruption alone, 15 by both, and 18 (22.2%) by neither. The determinants were predominantly chromosomal point mutations in AMRFinderPlus's catalogue rather than acquired genes: *pmrB* T157P in 10 isolates, *mgrB* nonsense and frameshift alleles in 12, *crrB* substitutions in 4, and single substitutions in *pmrA* (two isolates), *pmrB* S85R, *phoQ* and *lapB*, with only 9 isolates carrying any *mcr* gene. Correspondingly, the largest disrupted *mgrB* category here was premature stop (11 genomes), where internally it was the assembly-structural categories of §3.4. Point substitutions are what AMRFinderPlus detects and insertion-driven disruption is what it misses, so a collection weighted toward the former leaves the off-the-shelf tool less to miss and the targeted check less to add. The check still added 27.2 points over AMRFinderPlus alone, explaining 22 resistant isolates it missed, so the contribution of §3.3 replicates out of sample.

Two secondary results follow. The missense tier added 1.2 points externally (77.8% to 79.0%, on three isolates) against 10.6 points in-sample, independently confirming the decision in §3.5 to discard it. And 22.2% of external resistant isolates remain unexplained by any

determinant or *mgrB* disruption, close to the internal picture, so the unexplained fraction is a property of the mechanism catalogue rather than of one collection.

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4. Discussion

Three findings here bear on how genomic AMR prediction should be evaluated, and they point in the same direction: the limiting factor is not model capacity but the relationship between the training collection and the target population.

The *mgrB* result is the smallest of the three but the most concrete. A two-minute check of a single 47-codon locus recovers 28 points of colistin sensitivity over AMRFinderPlus — though only six over Kleborate, which already detects truncation, and we think the six-point figure is the one that should be quoted. The mechanism is instructive even where the increment is modest: the categories carrying most signal are not sequence features but assembly artefacts, contig boundaries that are diagnostic because an IS element has landed in a very short gene. Three independent lines of evidence support that reading (§3.4) — the breaks sit a median of 79 bp from *mgrB* against 80 kb in intact genomes, the adjacent sequence is multi-copy within its own assembly beyond what length-matched contig ends show, and 98% of these sequences recur in unrelated isolates. Screens that ask "which genes are present?" cannot see this; a screen that asks "is this locus whole?" can. That generalises to any resistance determinant whose loss, rather than presence, confers the phenotype.

The missense tier is a cautionary result we think worth reporting in full. The variants were individually plausible, several fell on structurally critical cysteines, and the in-sample gain was 19 points. Out-of-sample it was one. Nothing about the biology was obviously wrong; what was wrong was that the variants were chosen by looking at the same isolates used to score them. We suspect this failure mode is common in curated resistance-determinant catalogues assembled from observational association, and it is invisible to any validation scheme that does not group by study.

The heterogeneity result (§3.6) is in our view the most important, and the least avoidable. With nothing fitted, the same mechanism-specified rule achieved 39.5% sensitivity in one collection and 70.6% in another. Colistin resistance is simply constituted differently in different isolate sets. This bounds what any method can achieve: a model trained where *mgrB* explains 70% of resistance will underperform where it explains 40%, however well regularised. It also bears on the crossover in §3.7. We initially read that crossover as the models memorising lineage, but the sequence-type baseline in §3.8 shows otherwise: for colistin, type alone predicts barely better than chance. What the models lose when types are held out is therefore a finer-grained resemblance between related isolates, not type identity — a resource that likewise disappears on genuinely novel lineages, but one that a reader should not mistake for simple label memorisation.

We want to be careful not to overstate the comparison with machine learning. The rule leads under lineage-aware splitting by 0.028 and 0.050 balanced-accuracy points; both intervals exclude zero, but after correcting across the 54-comparison family only the second survives a Bonferroni threshold, and the first is better described as parity than as an advantage. For tigecycline the models win under every split. The defensible claim is not that rules beat models, but that the models' advantage under conventional validation is attributable to information that will not be present in deployment. Yu and colleagues demonstrated this across species using clade-based splits [1]; the contribution here is to show it at the level of a single identifiable mechanism, where one can point to exactly what the rule uses and what the model substitutes for it.

The label findings carry comparable methodological weight. Roughly a third of our usable labels did not exist as S/I/R calls but as uninterpreted MICs. Conversely, a substantial frac-

tion of apparent genotype error was *created* by harmonisation: the gentamicin borderline effect moves rule balanced accuracy by five points and depends entirely on whether one applies EUCAST or the CLSI revision under which those isolates were originally reported. A benchmark that silently selects or mixes standards reports a number whose provenance a reader cannot assess.

4.1 Limitations

Public surveillance data over-represent outbreaks and high-risk clones; two BioProjects supply 44% of our genomes, and prevalences here are not population estimates. Residual label error remains beyond the one flagged study. Colistin and tigecycline have modest numbers of resistant isolates (367 and 261). Collection year is missing for 34% of genomes and dated genomes concentrate in 2010–2014, so temporal validation was possible for three drugs only. Features are presence/absence calls from assemblies: copy number, expression, plasmid context and promoter-region insertions are not represented, and the 144 colistin-resistant isolates with intact *mgrB* coding sequence likely include promoter insertions we cannot see. Hyperparameters were fixed rather than tuned, so the models are reasonable defaults rather than optimised systems. Finally, while the positional, repeat-content and cross-genome evidence in §3.4 collectively supports the IS interpretation, we did not identify which element is involved or confirm insertion in any individual genome; that requires an IS reference database or long-read assemblies of these isolates, neither of which we generated. The mechanistic label on those categories therefore remains an inference, albeit a well-supported one. The external cohort did not advance this question: only one of its genomes fell into the assembled-insertion category, so the element remains unidentified there too.

The external cohort has limitations of its own. It is small (263 isolates, 81 resistant), so its sensitivity interval spans 18 points. Its isolates are those whose submitters chose to deposit colistin susceptibility results, which over-represents collections curated for characterised mechanisms; its higher sensitivity should therefore be read as an estimate on a mechanism-weighted collection rather than a field estimate, and the internal and external figures bracket rather than supersede one another. Thirteen isolates retain the submitter's own breakpoint interpretation, though excluding them changed sensitivity by 0.5 points. For colistin specifically the CLSI and EUCAST breakpoints effectively coincide, so harmonisation altered one label among the 355 records retrieved; it remains necessary for the other drugs but did little work here.

Most importantly, this study measures discrimination, not clinical usefulness, and the two come apart. Sensitivity and specificity are properties of the rule; predictive values also depend on how common resistance is in the population tested, and 24% of our colistin-tested genomes are resistant — a surveillance figure far above most clinical settings. Applying the internal rule's 56.7% sensitivity and 95.2% specificity at a 2% prevalence gives a positive predictive value of 19.4%: four in five genotype-positive isolates would not be resistant. At our own cohort's prevalence the complementary error dominates, with a negative predictive value of 87.4%, meaning 12.6% of genotype-negative isolates are resistant. For a last-line agent these two errors are not interchangeable — withholding colistin unnecessarily and administering it ineffectively carry different costs — and we performed no decision analysis weighing them. Nothing here establishes that genotypic prediction should inform colistin therapy, and we make no such claim: the rule's role in this paper is as a mechanism-specified baseline against which to measure validation design. Predictive values across a range of prevalences are tabulated in `reports/review/predictive_values.md`.

4.2 Recommendations

1. **Report a mechanism-specified rule baseline per drug.** It is the only way to tell whether a model has learned a mechanism or a lineage.

2. **Select determinants out-of-sample.** A plausible variant enriched in a cohort is not a validated determinant; leave-one-study-out selection cost our missense tier 18 of its 19 points.
3. **State and harmonise the breakpoint standard, and report sensitivity to borderline labels.**
4. **Use lineage-aware splits and report the gap from random splitting.** The gap was under one point where rules work and 11–15 points where they do not.
5. **Group on study and lineage together, not either alone.** In our cohort, study-aware splits still shared a sequence type with training data for 74–88% of test genomes, and lineage-aware splits still shared a BioProject for 87–95%. Temporal splits constrained neither (73–78% and 76–82%). Each single scheme leaves the other dimension almost entirely unconstrained.
6. **Report a lineage-only baseline.** It costs nothing and distinguishes a model that has learned a mechanism from one that has learned who its neighbours are.
7. **Report per-collection performance, not only pooled.** Pooled estimates concealed a two-fold range in our data, and a 21-point difference between our cohort and an external one.
8. **Verify species independently of the source's own labels.** Six of 270 isolates in NCBI's *K. pneumoniae* taxgroup were *K. aerogenes*, including complete genomes; Kleborate's species check caught them and AMRFinderPlus did not.

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5. Data and code availability

Code, pinned environments, provenance manifests (queries, timestamps, SHA-256 checksums), decision records, reports and figures are at https://github.com/omarfaruqe/klebsiella-amr. Derived tables regenerate deterministically from the documented pipeline. A model card is at `docs/model-card.md`. An archived snapshot of the code as submitted is deposited at Zenodo (version DOI 10.5281/zenodo.23634240; all versions: 10.5281/zenodo.23634239).

Internal genome assemblies and AST records are from BV-BRC. The external validation cohort (§2.10) derives from NCBI Pathogen Detection; the SQL that retrieved it is version-controlled with the code that consumes it, and the accession list, per-file checksums and harmonised labels are in the repository.

6. Author contributions

Following CRediT. **Md Omar Faruqe:** conceptualisation, methodology, software, formal analysis, data curation, investigation, validation, visualisation, writing — original draft. **Md Humayun Kabir:** conceptualisation, supervision, project administration, writing — review and editing.

7. Funding

This research received no external funding.

8. Competing interests

None declared.

9. References

All references were verified against the publisher record during preparation (2026-10-24).

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